AP Precalculus: 2026–2027

Unit 1: Polynomial and Rational Functions   Lesson 1.10: Rational Functions and Holes

Primary Objective: Students will identify holes in rational function graphs by locating shared zeros of the numerator and denominator where numerator multiplicity is greater than or equal to denominator multiplicity, simplify to find the hole's coordinates, and express the hole location using limit notation. *Skill: MP 3.C*

Priority Ideas & Skills

AP Skills	Key Understandings
MP 3.C — Support conclusions or choices with a logical rationale or appropriate data.	• A hole occurs at $x = c$ when the numerator and denominator share a zero at c with numerator multiplicity $m \geq$ denominator multiplicity n. (EK 1.10.A.1) • The y-coordinate of the hole is $L = \lim_{x\to c} r(x)$, found by substituting $x = c$ into the simplified form. (EK 1.10.A.2) • Both one-sided limits equal L: $\lim_{x\to c^-} r(x) = \lim_{x\to c^+} r(x) = L$.

Vocabulary, Concepts, & Theorems

hole in a graph	A missing point (c, L) where $r(c)$ is undefined but $\lim_{x\to c} r(x) = L$ exists.
removable discontinuity	Another name for a hole; the discontinuity is "removable" by redefining the function at one point.
common factor	A polynomial factor shared by numerator and denominator; canceling it reveals a hole.
multiplicity	The number of times a given zero appears as a factor.
limit notation	$\lim_{x\to c} r(x) = L$: the outputs of r approach L as inputs approach c.

Activate Prior Knowledge & Spiral Review

Spiral review (warm-up) covers: identifying vertical asymptotes from 1.9 (shared zero, numerator nonzero or denom mult > num mult); interpreting one-sided limit behavior at a VA; factoring a quadratic (bridge to factoring numerators and denominators for holes).

Hook

Suppose your GPS loses signal for exactly one moment — at $t = 5$ seconds — but your position function is still defined everywhere else. The output *at* $t = 5$ is missing, but we can figure out where it "should" be by looking at what the function approaches from both sides. This missing-but-predictable point is exactly what a hole is in a rational function.

Lesson

Part 1 (10 min): Holes vs. Vertical Asymptotes. Given a shared zero $x = a$ with numerator multiplicity m and denominator multiplicity n:

- $m \geq n \Rightarrow$ **HOLE** at $x = a$
- $m < n \Rightarrow$ **Vertical Asymptote** at $x = a$
- Zero only in denominator $\Rightarrow$ Vertical Asymptote

Connect to 1.9: VAs arise when denom is zero and numerator is nonzero *or* when $m < n$. Today's lesson is the complementary case ($m \geq n$).

Part 2 (15 min): Finding Hole Coordinates. Four-step procedure:

1. Factor numerator and denominator.
2. Identify shared zeros and their multiplicities.
3. Cancel the common factor.
4. Substitute $x = c$ into the simplified form to find L.

Example 1: $r(x) = \frac{x^2-4}{x-2} = \frac{(x-2)(x+2)}{x-2}$. Cancel $(x-2)$. Simplified: $x+2$. Hole at $(2, 4)$.

Example 2: $r(x) = \frac{x^2-1}{(x-1)(x+2)}$. Cancel $(x-1)$. Simplified: $\frac{x+1}{x+2}$. Hole at $\left(1, \frac{2}{3}\right)$. VA at $x = -2$.

Part 3 (5 min): Limit Notation. If the hole is at (c, L), then $\lim_{x \to c} r(x) = L$ with $\lim_{x \to c^-} r(x) = \lim_{x \to c^+} r(x) = L$. A hole is a *removable* discontinuity: $r(c)$ undefined, but limit exists.

Active Monitoring

- Cold-call after Part 1: "Why does $m \geq n$ produce a hole instead of an asymptote?"
- Watch for students who cancel factors but forget to check the *simplified* form for remaining VAs (e.g., Example 2 still has VA at $x = -2$).
- Confirm students substitute into the simplified expression, not the original.

Group Work & Differentiation

- **Tier R (Remediate):** Numerator/denominator pre-factored; classify hole vs. VA; fill table.
- **Tier A (Apply):** Factor independently; apply multiplicity rule; find coordinates (c, L).
- **Tier E (Extend):** Write limit notation; explain reasoning; construct a rational function with specified holes and VAs.

Individual Work & Assessment

Exit ticket (2 questions, 1 page): factor $r(x) = \frac{x^2-x-2}{x^2-4}$, identify shared zeros, classify each as hole or VA, find hole coordinates, and write the limit statement. Collect and sort by Tier R/A/E for formative data before Lesson 1.11.

Reinforcement & Extension

Homework: 5 problems covering identification of holes and VAs, multiplicity reasoning, writing limit notation in words, a higher-multiplicity case ($x^3 - x^2$ over $x^2 - x$), and an AP-style MC problem. Preview of Lesson 1.11: equivalent representations of rational functions — connecting factored, standard, and graphical forms.